

rely on hand-crafted rules with fixed parameters or limited adaptation. Neural networks instead learn complex, nonlinear relationships between observables and optimal sensor weights, thereby improving performance under challenging or uncertain conditions.

Implementing neural-network inference in radiation-hardened hardware remains demanding. Conventional graphics processing units used in terrestrial ML applications are not designed to withstand space radiation. As a result, current systems use radiation-hardened field-programmable gate arrays configured with custom neural-network accelerators.

Recent research explores spiking neural networks and neuromorphic hardware that process information via event-driven computation. These architectures promise substantial power-efficiency gains by activating processing only when events occur, enabling continuous low-power

monitoring that escalates to deeper diagnostics only when necessary.

Electronics in detail

Radiation tolerance continues to shape electronic design. Soft errors in memory and logic generate transient faults that can resemble sensor anomalies, so FDIR systems must distinguish permanent failures from brief disturbances. Designers use error-correcting codes, watchdogs, memory scrubbing, and filters that ignore short-lived glitches.

Power budgets enforce trade-offs between computation and sensing fidelity. High-rate star-tracker processing or large particle filters may be impractical. Instead, hardware-accelerated feature extraction reduces downstream processing, and asynchronous scheduling shifts heavy computation to low-activity periods.

Time synchronisation also remains critical. Distributed sensors require precise temporal alignment

to accurately fuse asynchronous measurements. Precision timing hardware, ground-derived time transfer, and sub-microsecond time-stamping ensure coherence.

Signal conditioning and interface circuits reduce noise and provide diagnostic telemetry. Because fusion and diagnosis algorithms assume specific noise models, poor conditioning degrades both accuracy and fault detectability.

Supervision, policy, and mission modes

A navigation system's autonomy must be governed by supervisory policies that define safe behaviour under faults. Graceful degradation allows continued operation with reduced precision, while safe-hold modes prioritise spacecraft survival over mission objectives.

FDIR communicates confidence metrics and recommended reconfiguration actions to the fusion layer. If a star tracker becomes compromised

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